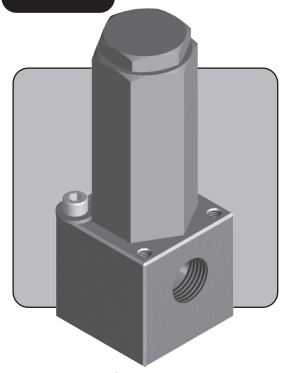
RL50-G

RELIEF VALVE

1/2″

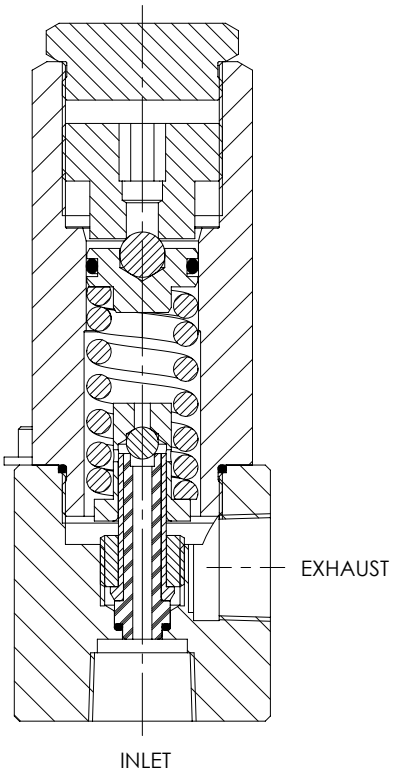
UP TO 414 BAR 6,000 PSI



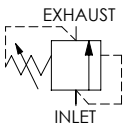
The RL50-G is a 1/2″ ported relief valve. Designed for emergency relief applications, it may be used to protect systems up to 414 bar (6,000 psi).

The valve is a proportional metering style relief valve with soft seats enabling its use with both liquids and gases, and is approved as a safety accessory to category IV of the Pressure Equipment Directive.

- Approved to category IV of the Pressure Equipment Directive (2014/68/EU)
- Available preset via internal adjustment or with an external handwheel adjustment option
- Wire locking and tagging available on request
- Minimal leakage at 90% of set pressure
- Reseat within 20% of set pressure
- Repeatabily within ±5% of set pressure
- Subsea variants available, contact us for advice
- Suitable for use with air, nitrogen, sweet natural gas, mineral oil, water glycols and plain water
- Suitable for many other media, contact us for advice
- Various spring ranges available



Specifications

BASIC MODEL NUMBER	RL50-G
SYMBOL	
MAX WORKING PRESSURE	414 bar (6,000 psi)
MAX OUTLET PORT PRESSURE	69 bar (1,000 psi) Note: Any outlet port pressure is directly additive to the set pressure
ORIFICE SIZE	Ø3.8mm (Ø0.150″)
FLUID	Liquids and Gases See materials section
TEMPERATURE RANGE	See Product Selector opposite and Technical Data section
PORT SIZE	1/2″
WEIGHT	1.75 kg (3.9 lb)

Specifications may change without notice

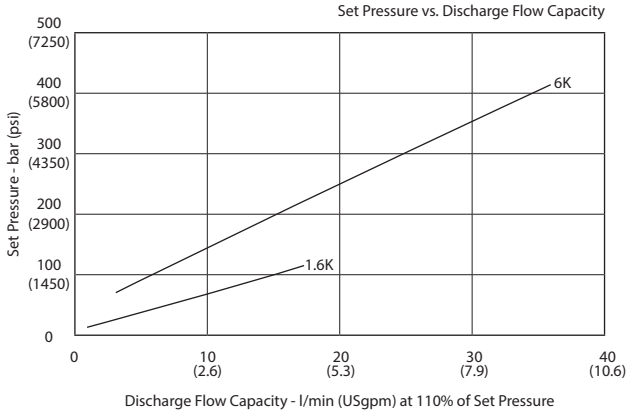
Materials

Externally Exposed Parts: 316 stainless steel on all pressure retaining components, and 300 series stainless steels on non-critical parts.

Internally Wetted Parts: 316, 302 and 431 stainless steel, PEEK and silicon nitride. The standard valve is designed for use with air, nitrogen, sweet natural gas, mineral oils, water glycols and plain water. Contact us for advice when use with other media is required.

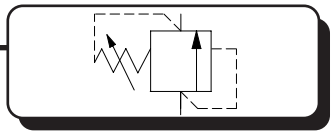
⚠ The standard valve has Viton® seals. Further seal options are available via the Product Selector. Compatibility with the working fluid at the operating temperature must be considered.

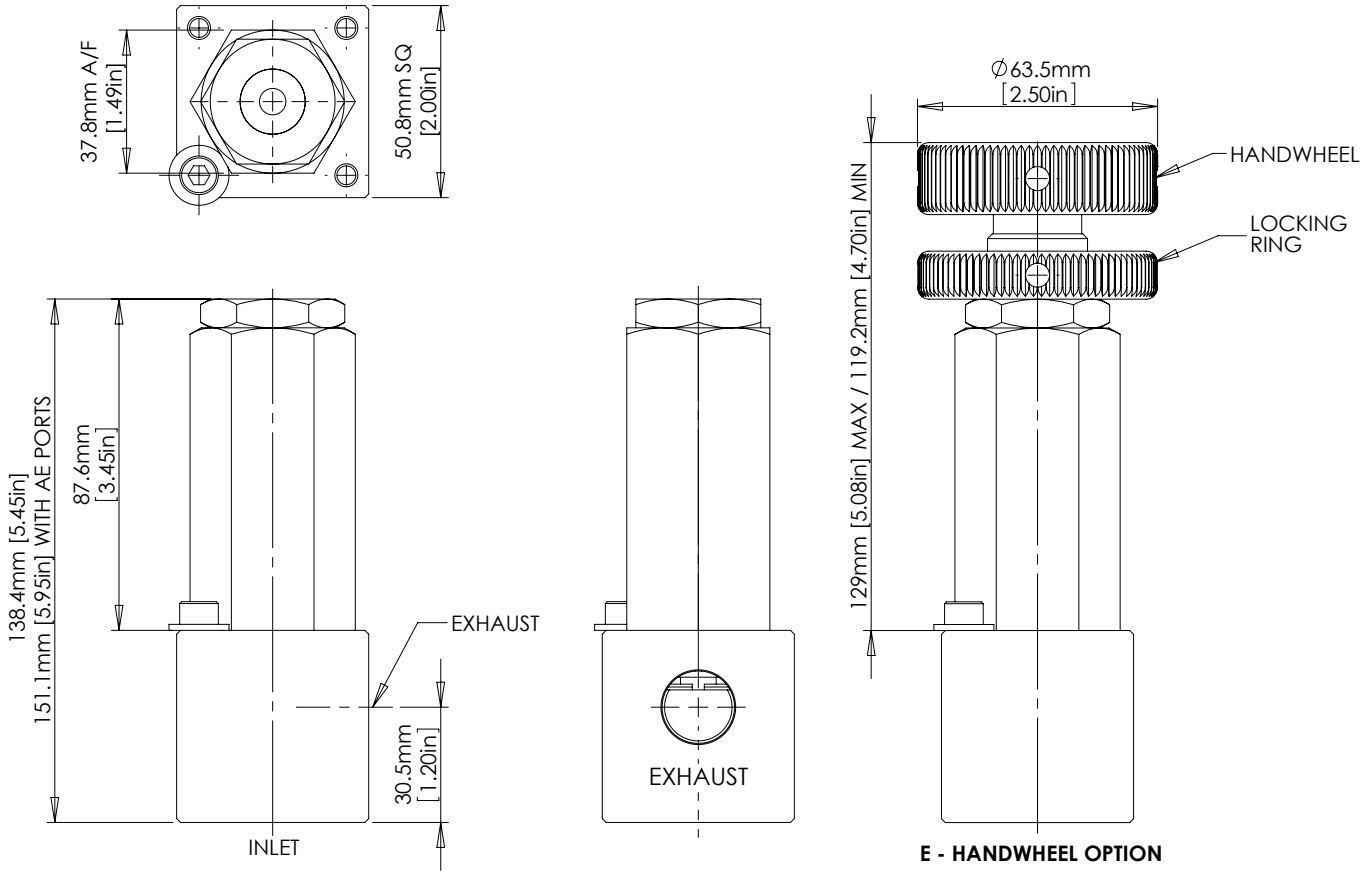
Typical Performance



Typical performance based on water
Contact us with system details for an application specific calculation

Installation Diagram





E - HANDWHEEL OPTION

Product Selector

RL50 - 56AE - 50P - E - G - NBR - 1.6K

VALVE TYPE	INLET PORT OPTIONS	EXHAUST PORT OPTIONS	VALVE OPTIONS	SEAT TYPE	SEAL OPTIONS	SET PRESSURE RANGE
RL50	P 1/2" BSPP female inlet and exhaust	Leave blank for P and N inlet options 50P 1/2" BSPP female exhaust	Leave blank if none required E Handwheel (external) adjustment	G option must be selected G Soft seats	Leave blank for Viton® seals -10˚C to +80˚C NBR Nitrile seals -30˚C to +80˚C	1.6K 13.8 - 110 bar (200 - 1,600 psi)
	N 1/2" NPT female inlet and exhaust	50N 1/2" NPT female exhaust	⚠ Exposure to high and low fluid temperatures will influence handle temperature. Operator safety must be considered		EP EPDM seals -40˚C to +80˚C	6K 69 - 414 bar (1,000 - 6,000 psi)
	56AE 3/8" OD medium pressure tube inlet				KLZ Kalrez® seals -25˚C to +80˚C	Advise us of set pressure if factory pre-setting is required
	81AE 9/16" OD medium pressure tube inlet				HNBR Hydrogenated Nitrile seals -20˚C to +80˚C	
					Further seal options are available on request	

Repair and seal kits are available on request

For a seal kit add SK at the end of the model code and for a repair kit add RK